```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
```

```
wine = load_wine()

X = wine.data
y = wine.target

feature_names = wine.feature_names
target_names = wine.target_names

df = pd.DataFrame(X, columns=feature_names)
df["target"] = y

df.head()
```

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	col
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	0.28	2.29	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	0.26	1.28	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	0.30	2.81	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	0.24	2.18	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	0.39	1.82	

```
print("Dataset shape:", X.shape)
print("Number of features:", X.shape[1])
print("Number of samples:", X.shape[0])
print("Classes:", target_names)
```

```
Dataset shape: (178, 13)
Number of features: 13
Number of samples: 178
Classes: ['class_0' 'class_1' 'class_2']
```

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   alcohol                       178 non-null    float64
 1   malic_acid                    178 non-null    float64
 2   ash                           178 non-null    float64
 3   alcalinity_of_ash             178 non-null    float64
 4   magnesium                     178 non-null    float64
 5   total_phenols                 178 non-null    float64
 6   flavanoids                    178 non-null    float64
 7   nonflavanoid_phenols          178 non-null    float64
 8   proanthocyanins               178 non-null    float64
 9   color_intensity               178 non-null    float64
 10  hue                           178 non-null    float64
 11  od280/od315_of_diluted_wines  178 non-null    float64
 12  proline                       178 non-null    float64
 13  target                        178 non-null    int64  
dtypes: float64(13), int64(1)
memory usage: 19.6 KB
None
```

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("Original shape:", X.shape)
print("Scaled shape:", X_scaled.shape)
```

```
Original shape: (178, 13)
Scaled shape: (178, 13)
```

```
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

print("Original dimensions:", X.shape[1])
print("Reduced dimensions:", X_pca.shape[1])
```

```
Original dimensions: 13
Reduced dimensions: 2
```

```
print("Explained variance ratio:")
print(pca.explained_variance_ratio_)

print("\nTotal variance retained:")
print(pca.explained_variance_ratio_.sum())
```

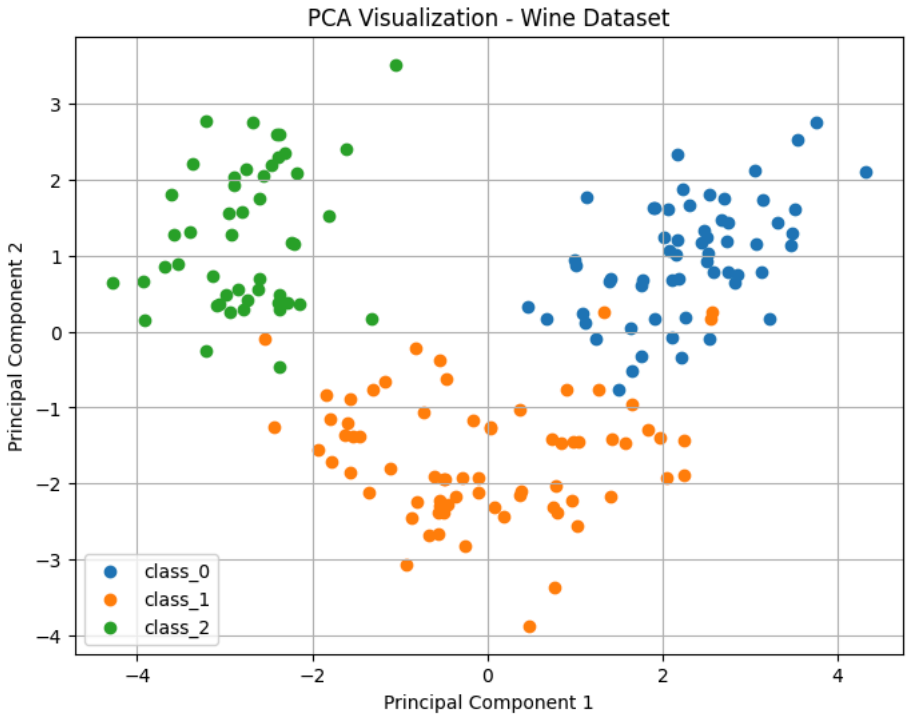
```
Explained variance ratio:
[0.36198848 0.1920749 ]

Total variance retained:
0.5540633835693526
```

```
plt.figure(figsize=(8, 6))

for i, target_name in enumerate(target_names):
    plt.scatter(
        X_pca[y == i, 0],
        X_pca[y == i, 1],
        label=target_name
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Visualization - Wine Dataset")
plt.legend()
plt.grid()
plt.show()
```




```
tsne = TSNE(
    n_components=2,
    perplexity=30,
    random_state=42
)

X_tsne = tsne.fit_transform(X_scaled)

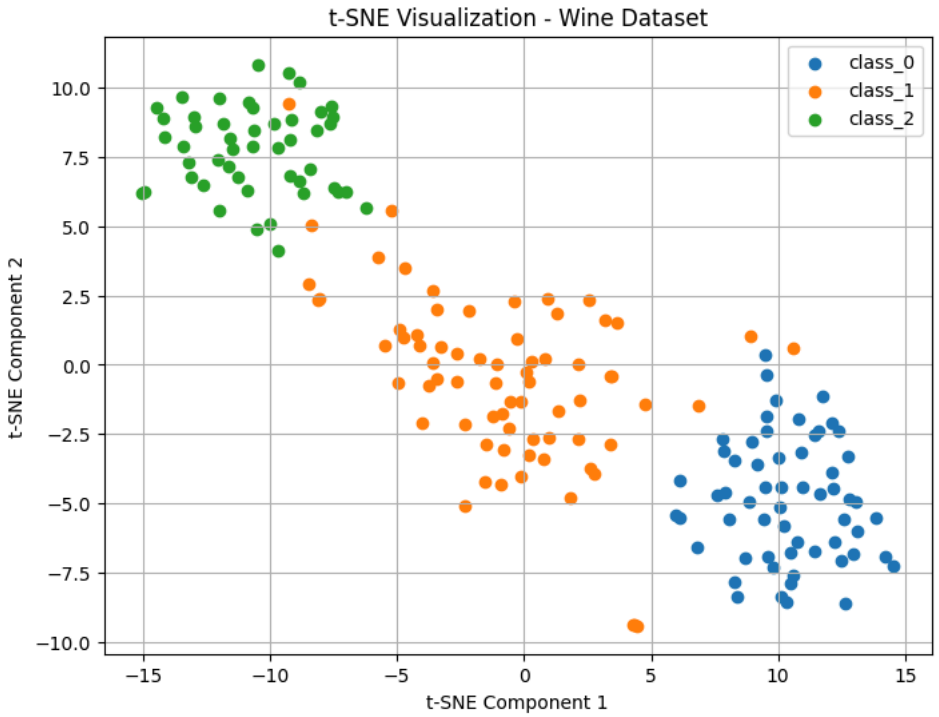
print("Original dimensions:", X.shape[1])
print("Reduced dimensions:", X_tsne.shape[1])
```

```
Original dimensions: 13
Reduced dimensions: 2
```

```
plt.figure(figsize=(8, 6))

for i, target_name in enumerate(target_names):
    plt.scatter(
        X_tsne[y == i, 0],
        X_tsne[y == i, 1],
        label=target_name
    )

plt.xlabel("t-SNE Component 1")
plt.ylabel("t-SNE Component 2")
plt.title("t-SNE Visualization - Wine Dataset")
plt.legend()
plt.grid()
plt.show()
```




```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# PCA
for i, target_name in enumerate(target_names):
    axes[0].scatter(
        X_pca[y == i, 0],
        X_pca[y == i, 1],
        label=target_name
    )

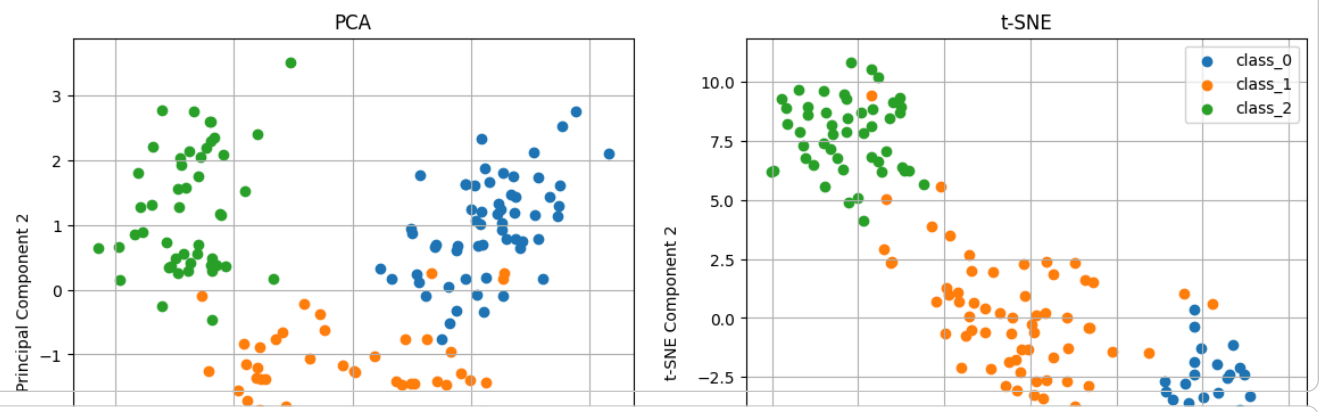
axes[0].set_title("PCA")
axes[0].set_xlabel("Principal Component 1")
axes[0].set_ylabel("Principal Component 2")
axes[0].grid()

# t-SNE
for i, target_name in enumerate(target_names):
    axes[1].scatter(
        X_tsne[y == i, 0],
        X_tsne[y == i, 1],
        label=target_name
    )

axes[1].set_title("t-SNE")
axes[1].set_xlabel("t-SNE Component 1")
axes[1].set_ylabel("t-SNE Component 2")
axes[1].grid()

axes[0].legend()
axes[1].legend()

plt.show()
```




```
comparison = pd.DataFrame({
    "Method": ["PCA", "t-SNE"],
    "Type": ["Linear", "Non-linear"],
    "Original Dimensions": [13, 13],
    "Reduced Dimensions": [2, 2],
    "Main Purpose": [
```